

Task: Divide the given network into equal-sized subnets using subnetting. Configure the PCs, switches, and routers with appropriate IP addresses. Connect the departmental networks using routers and implement static routing. Verify communication between PCs belonging to different subnets.

Question 4 – University Network: Casting & Network Communication

A university network consists of multiple PCs, switches, routers, and a server distributed across different

subnets. The administrator wants to demonstrate unicast, broadcast, and multicast communication.

Task: Design the network in Cisco Packet Tracer with appropriate IP addressing and subnetting.

Configure static routes between the networks. Using Simulation Mode, demonstrate and analyze unicast

communication between two specific PCs, broadcast communication within a subnet, and multicast communication to a selected group of devices. Record the packet path and explain how switches and routers handle each type of traffic.

